

Venkata Saratchandra Patnaik Markonda Patnaikuni

(602) 301-8122 | smarkond@asu.edu | <https://www.linkedin.com/in/saratchandra-92015>

EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ.

TECHNICAL SKILLS

Programming & Scripting Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C

GPU & Parallel Computing: CUDA Programming, PyCUDA, Shared Memory Optimization, Loop Unrolling, Tiling, Grid-Block-Thread Architecture

Web & Mobile Development: React.js, Node.js, Express.js, HTML/CSS, Bootstrap, Android Studio, Flask, REST APIs

Databases: MySQL, PostgreSQL, Google Firebase, MariaDB

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- **Led the end-to-end deployment of Capsequo**, a Python-based ML captioning pipeline using real-time API integration, reducing video stream latency by 36% and delivering scalable solutions for media clients.
- **Architected and managed 15+ microservices on AWS EKS** using Docker, Kubernetes, and ArgoCD, boosting deployment frequency by 30% and enhancing CI/CD reliability through GitOps automation and rollback strategies.
- **Developed automation and monitoring tools with Python and Terraform**, reducing customer onboarding time by 90% and video delivery failures by 95%; conducted code reviews and mentored interns to drive quality and system resilience.

Web Development Intern: Blueplanet Solutions Inc., India

April 2021 - June 2021

- **Optimized MySQL database** using stored procedures for 'Campus Club', improving query speed by **50%** and enabling faster data access for 1,000+ student profiles.
- **Enhanced a responsive student search UI** with HTML, CSS, JavaScript, and PHP, reducing search time by **60%** and improving user experience.
- **Analyzed web server logs** to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Parallel Implementation of Gaussian Filter on GPU (CUDA, PyCUDA, Image Processing)

- Implemented 2D Gaussian convolution on GPU using CUDA & PyCUDA, optimizing performance with thread grids, blocks, and shared memory.
- Applied techniques like loop unrolling, tiling, and separable filters, achieving up to 20× speedup over a CPU-based MATLAB implementation.
- Validated results using PSNR and SSIM metrics; analyzed computation time for various kernel/image sizes to quantify performance gain.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Developed an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda; automated deployments via shell scripts and passed 100+ real-time autograder test cases.